

CS728 Programming Assignment 3

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April 5, 2026

1 Part 1: Classical Retrieval

1.1 Comparison of Retrieval Methods

Method	Recall@1	Recall@5
BM25	0.2312	0.4114
msmarco-MiniLM	0.3882	0.6106
UAE-large-v1	0.6502	0.8842

Table 1: Recall at different ranks for the three retrieval methods.

2 Part 2: Lost-in-the-middle

2.1 Results

- Recall@1: 0.0122
- Recall@5: 0.2486

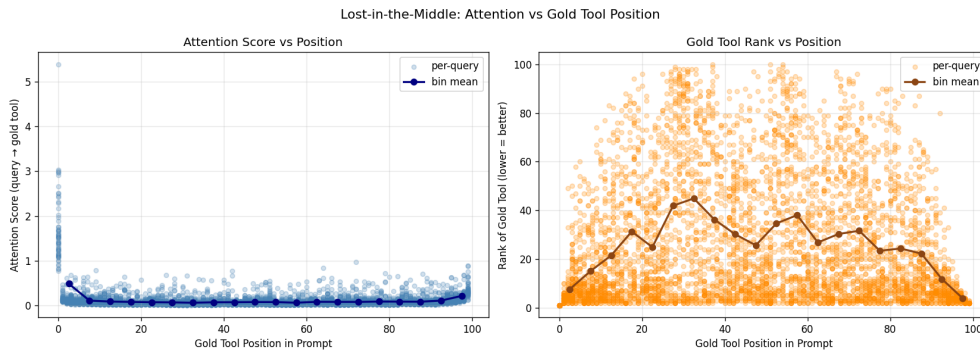


Figure 1: Position Effect on Attention Based Ranking

2.2 Plot

3 Part 3: Retrieval Heads

3.1 Head selection strategy

We use the Mean Reciprocal Rank (MRR) strategy to select heads. We score each (layer, head) pair based on its consistency in ranking the gold tool. Tools ranked lower have lower reciprocal score hence if the gold tool is ranked as the best consistently, that (layer, head) pair would have a score closer to 1. This way the top K heads are identified (20 in this case). The following heads were selected:

Selected Heads: [(6, 11), (7, 12), (4, 16), (5, 11), (8, 19), (10, 9), (2, 20), (10, 1), (6, 8), (10, 23), (6, 9), (5, 10), (8, 20), (8, 13), (12, 13), (5, 9), (10, 13), (10, 11), (8, 17), (10, 26)]

3.2 Results

- Recall@1: 0.2632
- Recall@5: 0.5918

3.3 Comparison with previous methods

We see that choosing only the top 20 heads performs much better than aggregating over all heads. Hence we see that relevant information is mostly present in the top k heads. This approach performs similarly to msmacro-MiniLM from part 1.